



Determination of Silicon in Cast Iron and Steel



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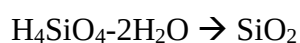
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Introduction:

The elemental composition analysis is a fundamental aspect of characterization, especially in steel and cast iron. Si plays a significant role in these materials, especially in controlling hardness, strength, and corrosion resistance. In cast irons, silicon acts as a graphitizing agent, while in steels, it functions primarily as a deoxidizer during manufacturing.

This experiment aims to quantitatively determine the Si content as a percentage in steel using a gravimetric method. The procedure involves dissolving the steel sample in an acid mixture (stock acid), followed by precipitation and ignition of silica (SiO_2).



↳ orthosilicic acid.

From the weight of SiO_2 , the Si percentage can be calculated using this formula:

$$\% \text{Si} = \text{weight of precipitate in 1gm} \times (28/60) \times 100$$

Apparatus:

- Steel Sample
- 41mL Stock acid (36mL HNO_3 + 30mL H_2SO_4 + 180mL H_2O)
- Beaker
- Funnel
- Filter paper
- HCl
- Hot water
- Crucible
- Furnace
- Conical flux
- Glass rod
- Weight machine

Experimental Procedure:

1. First, we took a 2g steel sample.

2. We mixed it with stock acid (41 mL) in a 400 mL beaker.
3. Then we evaporated it until white fumes appeared.
4. After cooling it, we added 30 cc concentrated HCl with 70 mL distilled water.
5. After that, we heated and boiled it for 2–3 minutes.
6. We then set up a conical flask, funnel, and pulp on it.
7. We filtered the sample with a glass rod and continued until the liquid over the pulp became clear. (During this process, we cleaned the beaker with hot water to remove any remaining solution.)
8. After that, we transferred the precipitate and pulp into a crucible and ignited it at 700°C.
9. Finally, we weighed the pulp with the crucible and then weighed the empty crucible (after removing all the pulp from it).



Figure 1: Stock Acid mixture with sample



Figure 2: As the time being, solution changes its color.



Figure 4: Filtering through Pulp Set-up



Figure 3: Crucible

Calculation and Result:

Crucible with precipitate, $W_{\text{ppt}} = 31.96$

Only Crucible weight, $W_{\text{crucible}} = 31.94$

Weight of precipitate with pulp ash, $W_{\text{pwa}} = 31.96 - 31.94 \text{ gm} = 0.02 \text{ gm}$

We know, Weight of pulp ash = 0.0014 gm

Weight of only precipitate = 0.0186 gm

So, 2gm steel has 0.0186gm SiO_2

1gm will have 0.0093gm

So, 1gm steel sample will contain 0.0093gm SiO_2

$$\begin{aligned}\% \text{Si} &= \text{weight of precipitate in 1gm} \times (28/60) \times 100 \\ &= 0.0093 \times (28/60) \times 100 \\ &= 0.434\end{aligned}$$

Hence, our steel sample has 0.434% Si.

Discussion:

1) Why was HCl used, not HF?

Here, we not only get orthosilicic acid as our precipitate, we also get Fe_2O_3 / FeO and also other sulphate, nitrate precipitates.

Here, HCl will react with these precipitates (except SiO_2) and dilute it into solution. For e.g.



But in the case of HF, it will also dissolve SiO_2 , which would give improper Si content in our calculations.



2) Why will not H_2SO_4 be added before, for alloy steel?

Alloy steel has lots of carbide formers (Ti, W, Mo, Cr). H_2SO_4 can not dissolve carbide from steel. However it will attack substitutional solid solution (in matrix) first and will form precipitate. For that reason we will not get proper evaluation from our collected precipitates.

That's why HNO_3 acid add first, then H_2SO_4 / stock acid.

3) Role of Si in steel and the best conditions for our sample:

- Si has a very low hardenability factor compared to other alloying elements.

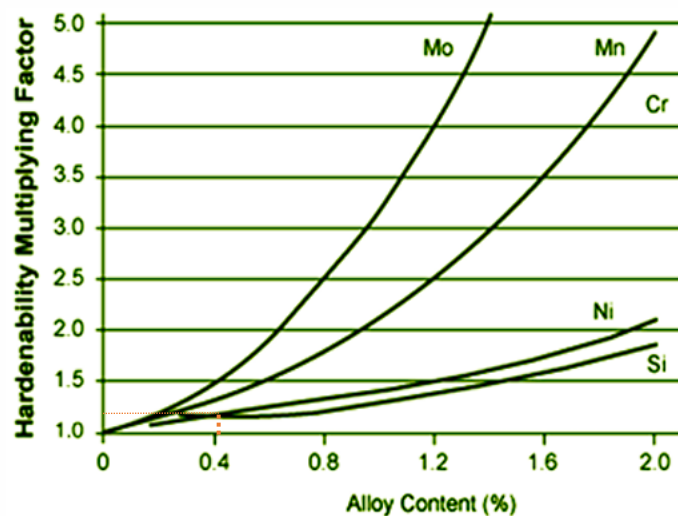


Figure 5: Hardenability Multiplying Factor of Different Alloying Elements

Our sample has hardenability factor = 1.2

- Si is a graphitizer. So, we can use our sample steel in oil hardening graphitic tool steels and air hardening graphitic tool steels.
- Si lowers the eutectoid carbon content but raises the eutectoid temperature. It has a detrimental effect on temper embrittlement.
- Si reduces impact toughness and raises the DBTT.

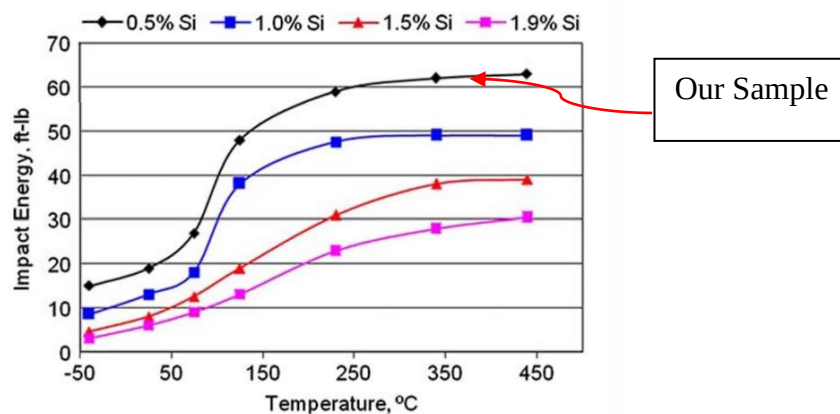


Figure 6: Effect of Si content on DBTT of steel

From the curve, I can say that we have good DBTT compared to other samples. That implies that our steel can work at high temperatures and will show a ductile nature. On the contrary, its properties will deteriorate at low temperatures.

- For best welding purposes, %Si < 0.1 and for better galvanizing purpose, it should not exceed 0.04% . So our steel sample is not applicable in welding and galvanizing coating.

4) Presenting all groups data:

Group	Crucible with Precipitate (g)	Crucible (g)	Precipitate with Pulp Ash (g)	Precipitate Only (g)	SiO ₂ in 1g Sample (g)	%Si
1	36.72	36.70	0.02	0.0186	0.0093	0.434
2	36.60	36.58	0.02	0.0186	0.0093	0.434
3	31.96	31.94	0.02	0.0186	0.0093	0.434
4	36.36	36.32	0.04	0.0386	0.0193	0.901
5	39.43	39.39	0.04	0.0386	0.0193	0.901
6	38.53	38.51	0.02	0.0186	0.0093	0.434
7	44.26	44.23	0.03	0.0286	0.0143	0.667
8	40.48	40.47	0.01	0.0086	0.0043	0.201
9	29.59	29.55	0.04	0.0386	0.0193	0.901
10	31.17	31.14	0.03	0.0286	0.0143	0.667
11	36.19	36.17	0.02	0.0186	0.0093	0.434
12	45.79	45.78	0.01	0.0086	0.0043	0.201

From this data table, Si is higher in group-4,5 and 9 's samples. And lowest in group- 8 & 12. Showing this in a bar graph:

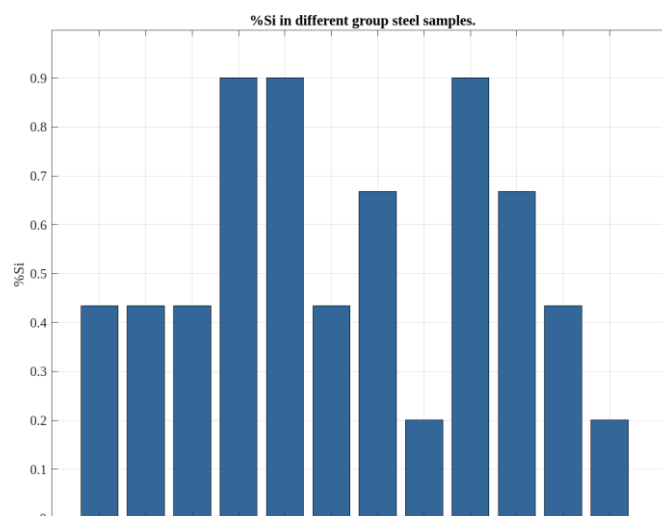


Figure 7: %Si in different groups